

INTERNATIONAL TRADE LIBERALIZATION AND DOMESTIC INSTITUTIONAL REFORM: EFFECTS OF WTO ACCESSION ON CHINESE INTERNAL MIGRATION POLICY

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Abstract—Economic institutions that impede factor mobility become more costly when an economy experiences substantial transitions such as trade liberalization. I study how trade triggers changes in labor institutions that regulate internal migration in the context of China's Hukou system. Using a newly collected dataset on prefecture-level migration policies, I document an increase in promigrant regulations following WTO entry and estimate the impact of prefecture-level tariffs on exports on migration regulations from 2001 to 2007. I find that regions facing more export market liberalization enacted more migrant-friendly regulations.

I. Introduction

TRADE liberalization is an important economic force driving changes in institutions. For example, Atlantic trade improved the protection of property rights (Acemoglu et al., 2005), Medieval trade in Venice pushed for constraints on the executive and innovations in contracting institutions (Trefler & Puga, 2014), modern import competition in the United States and Germany impacted electoral outcomes (Che et al., 2016; Jensen et al., 2017; Dorn et al., 2020; Dippel et al., 2021), and removal of quota restrictions on Chinese exports reduced misallocation resulting from the distorted quota-allocation (Khandelwal et al., 2013). Trade liberalization usually has an unequal impact across firms, industries, and regions, incentivizing factor reallocation. Thus it increases the effective cost of maintaining economic institutions that impede such reallocation, potentially leading to institutional reforms.

In this paper, I study how international trade liberalization affects institutions that regulate labor mobility in the context of China's Hukou system. The Hukou system is a vestige of a central-planning economy where the government creates artificial barriers between citizens in different geographic locations and different sectors. Before the economic reform in 1978, it was used to ration the allocation of all economic resources: land, jobs, goods, and social benefits. Even in the market-oriented economy today, people who live and work outside their Hukou prefecture need to obtain temporary registration to achieve a legal migrant status. But

all migrants, even legal ones, have access to a diminished level of public goods, such as medical insurance and public schools. The Hukou system makes internal migration across regional borders similar to international migration across national borders.¹ Despite fast economic growth in the 1990s, China's Hukou system remained rigid. However, around the time of WTO accession in 2001, the Chinese central government allowed prefecture-level governments to make their own Hukou regulation changes. A large body of literature has documented the profound impact of China's accession to the WTO on both the world economy and the Chinese economy.² I argue that the reduction in trade costs and the growth in Chinese export opportunities affected firms and consumers not only directly, through prices, but also indirectly, through changes in institutions. The economic costs of maintaining a rigid Hukou system could become very high when international trade opens up. Economic entities, in this case Chinese local governments, had the incentive to relax the Hukou restrictions and liberalize internal migration, particularly when a more flexible domestic labor market allowed government officials to reap larger gains from trade liberalization.

I identify the effect of trade liberalization on migration policy changes at the sub-national level. Identifying these effects at the country level is difficult for two reasons. At the country level, trade policies and migration policies are usually determined simultaneously, since countries with faster economic growth might choose both a more open trade policy and a more flexible migration policy. In addition, although trade shocks are relatively easy to measure, it is difficult to uniformly quantify migration regulations across different countries. The Chinese context has several unique features that enable me to avoid the simultaneity problem and solve the measurement problem. First, by receiving most-favored-nation (MFN) status after the WTO accession, tariffs on Chinese exports fell, and export growth followed. This overall decline in tariffs on exports affected regions within China differently, depending on their initial local industrial composition. I use these differential regional declines in tariffs on exports to identify the effect of trade.

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¹Similar systems also exist in other countries: *propiska* in Belarus, Kazakhstan, Russia, and Uzbekistan (abolished in Ukraine in 2001), *ho khau* in Vietnam, and *hoju* in North Korea (abolished in 2008). In 2016, 22.5% of the world population is subject to such internal migration regulations.

²See Autor et al. (2013), Pierce and Schott (2016), Amiti et al. (2017), and Handley and Limão (2017) on the impact of Chinese export growth on the U.S. economy, and Brandt et al. (2017), Facchini et al. (2019), Zi (2020), and Erten and Leight (2021) on the WTO accession on Chinese productivity, interregional migration, and intraregional structural transformation, among others.

Second, given the decentralized nature of Hukou reform in the 2000s, I collect a new dataset on prefecture-level migration regulations from 1978 to 2016 to measure prefecture-level migrant friendliness.³ I use a difference-in-difference identification strategy where I compare the change in migrant-friendliness of prefectures that experienced a big versus a small decline in tariffs on exports in the post-WTO period of 2001 to 2007. These two types of prefectures had similar migrant-friendliness in the pre-WTO period of 1995 to 2001. I find that liberalized trade policies, which increased demand for exports, led to relaxation of migration restrictions.

In addition to hand-coding the migrant-related regulations to evaluate whether they are promigrant or antimigrant, I also apply natural language processing methods with machine learning algorithms to validate the manual coding. From 2002 to 2007, 179 prefectures out of 340 enacted 748 new regulations related to migrant workers, addressing issues including vocational training, wage-arrears prevention, access to social insurance, and migrant children's access to local public schools.

I estimate the effect of changes in tariffs on exports on migration regulations across Chinese prefectures from 2001 to 2007 using a long-difference specification. I identify regional changes in tariffs on exports using a standard methodology; however, I use changes in tariffs on exports (similar to the that in Bustos, 2011; McCaig & Pavcnik, 2018; and Bombardini & Li, 2020), instead of tariffs on imports, which are more common in the literature.⁴ I calculate a prefecture's exposure to changes in tariffs on exports using the interaction of industry-level tariff reductions and prefecture-level industry employment shares. To address the concern that industry-level post-WTO tariff changes might be correlated with pre-WTO industry characteristics, I show that industry-level post-WTO tariff declines on exports were not correlated with pre-WTO export growth or tariff declines. The tariff reductions came from countries that import Chinese goods and should not be correlated with prefecture-level economic conditions. Empirically, I show that prefecture-level post-WTO tariff declines were not predicted by pre-WTO economic growth levels.

Overall, I find that prefectures that faced larger increases in access to export markets had larger increases in their migration regulation index, indicating more favorable treatment of migrants. The regulation score of prefectures whose declines in tariffs on exports were in the upper third of the distribution rose by 0.40 standard deviation more than the regulation score of prefectures in the lower third. Further, prefectures with a higher demand for migrants responded more positively to the decline in tariffs on exports. I also find that other WTO-induced export demand shocks generated similar regulatory changes, including (i) permanent nor-

mal trade relations (PNTR) shocks as in Pierce and Schott (2016) and Handley and Limão (2017), and (ii) Multi Fibre Arrangement (MFA) quota reduction as in Khandelwal et al. (2013). These results are robust to using trade measures following Autor et al. (2013) and using alternative measures of regulation scores.

I complement the main analysis with two additional exercises to show that these regulatory changes were associated with improved migrant well-being and increased migrant flows. The documented regulation changes are *de jure*. To provide some evidence on *de facto* regulation changes or the enforcement of the regulations, I show that in the 2005 cross-section, migrants in prefectures with higher migrant friendliness had higher levels of well-being, conditional on similar measures of well-being of locals. The well-being measures include migrant population's access to social insurance, wages, contract lengths, and children's access to local schools. This suggests that the migrant-friendliness score indeed captures the implementation of regulations that improve migrant well-being. I also show that changes in tariffs on exports led to changes in migrant flows, and these migrant flow responses were associated with the regulatory changes.⁵

This paper contributes to several literatures. Among the already-mentioned papers that study the determinants of institutions and the effects of the WTO, this paper complements Khandelwal et al. (2013) by providing the direct evidence on the impact of trade on observed regulation changes, instead of inferring institutional reforms from observed trade patterns. This method of direct measurement of institutional changes using regulation texts is in line with the recent political economy literature using text-based data (as in Gentzkow et al., 2019b and summarized in Gentzkow et al., 2019a).

A few theoretical papers study the interactions between trade liberalization and migration liberalization: specifically, how welfare gains from trade liberalization are affected by labor market frictions or how the effect of migration liberalization is compromised by the existence of trade frictions.⁶ Tombe and Zhu (2019), Fan (2019), and Ma and Tang (2020) focus on China, studying the aggregate and distributional effects of international and domestic trade on productivity where labor market frictions exist. While this literature takes labor market frictions as given, I argue that these labor market institutions are endogenous and respond to economic incentives. I provide empirical evidence on how a plausibly exogenous decline in tariffs on exports on regulations affect labor market frictions. To my knowledge, there is no study

³The measure is similar to that in Besley and Burgess (2004).

⁴The literature on the import competition effect of trade includes Autor et al. (2013), Kovak (2013), and Topalova (2010).

⁵It is difficult to separately identify the direct impact of changes in tariffs on exports on migrant flows through prices, and the indirect impact through regulation changes. See discussions in appendix C.11.1 on the threat to identification and various attempts to identify the causal impact of migration regulations on migrant flows.

⁶See Alessandria and Delacroix (2008), Kambourov (2009), Helpman and Itzhakovi (2010), and Caliendo et al. (2017).

that has documented the causal effects of trade liberalization on migration regulations.⁷

This study relates to the literature on fiscal competition (e.g., Fajgelbaum et al., 2019 and summarized in Oates, 1999 and Wilson, 1999). I show that regions compete to attract a common labor force by providing amenities or subsidies. However, I do not directly address efficiency issues related to such competition. Actually, when other distortions exist in the economy, this competition for labor could be welfare-improving for all. I discuss this possibility in section IX. Lastly, this paper is related to the literature on the effects of migration on economic outcomes (e.g., Card, 1990, 2001; Borjas, 2003; and Ottaviano & Peri, 2012, among many others). While most of the papers in this literature use exogenous increases in migrant flows, I emphasize the importance of regulatory forces in driving changes in migration.

The rest of the paper is organized as follows. In section II, I discuss the background on the Hukou system. Section III describes my new data on Hukou regulation changes. Section IV presents the declines in tariffs on exports associated with China's WTO accession. Section V discusses sample restrictions. Section VI presents the main empirical results on the effects of trade on migration policies. Section VII provides supplementary empirical evidence showing that these regulatory changes affected migrant well-being, and section VIII shows the effect of trade and regulation changes on migrant flows. Section IX concludes.

II. Background: The Hukou System in China

China's Hukou system is the internal registration system for Chinese citizens. Each individual has a Hukou status associated with a location and a sector (agricultural vs. nonagricultural) based on parents' status. To formally switch sector or prefecture, an individual needs to obtain a temporary Hukou registration enabling legal migrant status. Illegal migrants face the risk of retention and repatriation. Illegal migrants usually work on temporary jobs; formal manufacturing positions usually require temporary registration. Government jobs, jobs at state-owned enterprises, and many other permanent jobs are available only to locals. Even legal migrants with temporary Hukou suffer diminished access to public services such as medical insurance and public schools (CECC, 2005; Yusuf & Saich, 2008).

⁷Feler and Senses (2017) shows that trade shocks from China affect the local provision of public goods in the United States through the tax revenue channel. My paper adds to the discussion in two ways. First, local governments in Feler and Senses (2017) do not make adjustments in tax rates, and the public provision is a mechanical function of the tax revenue. In my case, the local government has the power to pass new regulations to adjust the amenity provision in response to trade shocks. In addition, the migration regulations control the number of migrants indirectly through the amenity level, and migrant flows can affect future economic growth. Second, the migration regulation changes are part of the Hukou reform, increasing labor mobility within China, and potentially decreasing spatial labor misallocation.

Before 2000, the central government held a rigid stand on the Hukou system, and lower-level governments were universally subject to the national policy. It was difficult for an urban resident to get a Hukou in other prefectures, unless he or she found an official job in an urban area that sponsored Hukou changes. The process was even harder for those wishing to switch from agricultural to nonagricultural Hukou. There were tight annual quotas, most of which were assigned to people whose spouse held a nonagricultural Hukou.

The Hukou system has been linked to spatial disparities in income (Wang & Zuo, 1999; Tombe & Zhu, 2019). In 2000, 11% of the population was employed in a prefecture-sector other than their assigned Hukou. Migrant workers worked and lived under inferior conditions; their legal rights at work were not protected and they had limited access to local schools and hospitals.⁸

Around 2000, the central government started to soften its stance on issues related to Hukou. The Tenth Five-Year Plan talked specifically about reducing political barriers to migration.⁹ In addition, local governments were allowed some discretion to design their own reforms following the central government guidelines.¹⁰ The timing of the reform coincided with WTO accession; in research articles and interviews with government officials, WTO accession was described as a chance to reform the Hukou system.¹¹

The central government's evolving stance spurred substantial local responses. Prefectures started to improve the well-being of legal migrants and set up a pathway for some migrants to get permanent local Hukou. They set up guidelines to protect migrant workers' legal rights in the workplace and also granted partial access to the social safety net and other local amenities. Some prefectures allowed migrant children to enroll in local primary and secondary schools. A few prefectures established a point-based system for applying for a local Hukou.¹² Although migration was still regulated, the number of migrants increased. By 2010, the number of Chinese migrants was 260 million, almost doubling the 2000 figure, and a larger share of migrants was moving between prefectures.

⁸Source: http://www.gov.cn/zhuanti/2015-06/13/content_2878968.htm.

⁹From the Tenth Five-Year Plan: "We will adapt to the market-oriented employment mechanism . . . to have an orderly and reasonable allocation of rural and urban labor." Source: www.people.com.cn/GB/shizheng/16/20010318/419582.html, or www.gov.cn/gongbao/content/2002/content_61966.htm.

¹⁰According to a 2001 document by the State Council of China, "Local governments should take into consideration local economic and social development levels and conduct reforms that balance population growth, infrastructure, employment and social security, and other welfare programs." Source: www.gov.cn/zhengce/content/2016-09/22/content_5110816.htm.

¹¹An interview with the Minister of Public Security, Division of Hukou Management, in 2001 writes: "The employment system, education system, and social security system are all evolving, and it is about the time to partially liberalize internal migration. Entering the WTO is an opportunity to change the Hukou system from management to service"

¹²This is similar to the point-based system for immigration in Canada and Australia. Source: <https://www.canada.ca/en/immigration-refugees-citizenship/services/immigrate-canada/express-entry/eligibility/federal-skilled-workers/six-selection-factors-federal-skilled-workers.html>

III. New Data on Hukou Policies

A. Regulation Data and Manual Coding of Migrant Friendliness

To document the change in Hukou regulations, I collected government regulations from the website www.pkulaw.com. The website collects regulations from official government websites, government gazettes, repositories of laws and regulations, as well as documents provided by relevant government units. Through December 31, 2016, the database on local government regulations contains at least one regulation from 332 of China's 340 prefectures, and large cities such as Shanghai, Beijing, and Chongqing have more than 25,000 regulations. To my knowledge, this is the most comprehensive dataset on Chinese government regulations.¹³

I uncover and rate all the regulations that potentially affect the well-being of migrant workers, by either changing their income or access to amenities, or giving them access to local Hukou (which will indirectly provide income and amenities). First, to find the regulations that potentially affect migrant well-being, I search for regulations with the following words in their titles: *non-Hukou population*, *migrant worker*, *temporary residence*, and *Hukou*. *Non-Hukou population* and *migrant worker* are common terms used to refer to internal migrants in China. Since the very definition of the internal migrants is tied with Hukou status, *temporary residence* and *Hukou* are used to capture the regulations related to Hukou issues. Overall, the keyword search identified 4,273 migrant-related regulations issued from 1978 to 2016, where 1,358 were issued by province-level governments and 2,915 were issued by prefecture-level governments.

Second, I classify the regulations into different topic categories, based on the main issue they address: (1) work-related, covering topics such as ensuring timely wage payments, requirement on signing contracts with migrant workers, labor unions, vocational training, legal assistance, and other support services; (2) social-insurance-related, covering topics such as access to the social safety net, for example, migrant workers' access to unemployment insurance, injury insurance, medical insurance, and pensions; (3) local-public-good-related, regarding migrant children's access to local primary and secondary public schools, availability of health benefits such as free vaccination, housing support, and financial services; (4) Hukou-related, on administrative issues regarding temporary Hukou registration, including the fee for obtaining temporary registration; (5) family-planning-policy related, on the implementation of the one-child policy, and (6) miscellaneous issues.

Figure 1 provides a glimpse of contents of the regulations, using all 4,273 regulations. Panel a shows the top 200 words in the pool of regulations on work-related issues, with a larger font representing a higher frequency. We can

see that the important words include "(wage) arrear," "training," "professional," "project payment," "labor contract," "pay," and "enterprise," pointing to the issues encountered by migrant workers at their workplace. Panel b presents the most frequent words for the regulation on local-public-good-related topics. The important words are quite different from panel a, including "children," "school," "compulsory education," "vaccination," and "immune."¹⁴

Third, to evaluate the migrant-friendliness of the regulations, I create a five-point index with scores ranging from -2 to 2 , where -2 is strongly against migrants, and 2 is strongly promigrant. Note that within a topic category, there can be promigrant regulations and antimigrant regulations. I follow the following principles when rating a regulation.

1. In the work, social insurance, and local public good categories, regulations mentioning providing migrants and migrant children with access to workplace rights protection, social insurance, local public goods, and amenities are coded as positive, and the ones mentioning depriving or restricting access are coded as negative.
2. In the Hukou topic, regulations mentioning reduction of fees for temporary registration and simplification of procedures are coded as positive, and the ones with increases in fees are coded as negative. The regulations that provide minor, nonsubstantive revisions on temporary registration requirements are coded as 0.
3. In the family-planning topic, regulations that mention providing healthcare services to pregnant migrant women and free vaccinations to children are coded as positive. Otherwise, regulations are rated as 0, since people are subject to the one-child policy both in their home regions and in the regions where they live temporarily, and it is not clear which rules are more strict.
4. The purely administrative regulations (e.g., informing the logistics of getting certain documents, certificates, or proofs) are rated as 0.
5. To distinguish -2 versus -1 , and 2 versus 1 , the coding rules are as follows: (1) If the regulation is about setting up a complete, executable *guideline* for one specific issue, a score of 2 is assigned (-2 if it is against migrants). For example, some regulations require construction firms to set aside deposits before the start of the project, designated for payment of wages to migrant workers. Some regulations require firms to purchase injury and medical insurance for migrant workers. These regulations are likely to be in place for a long time and affect migrant well-being substantially. (2) If the regulation addresses one issue, but is more about *enforcement* of specified rules, a score of 1 is assigned (-1 if it is against migrants). For

¹³The dataset has also been used by Fan (2019) and Sun et al. (2011) to evaluate Hukou reforms. However, they only evaluate the regulations that allow changes in the Hukou status, while I include all regulations related to migrant well-being.

¹⁴Appendix A.3 shows the corresponding word clouds for the other topics. Note that the original word cloud is produced in Chinese, and synonyms in Chinese can translate into the same English words, resulting in the same words showing up more than once in the English-version word cloud.

FIGURE 1.—THE MOST FREQUENT WORDS IN REGULATIONS ON WORK-RELATED ISSUES AND LOCAL-PUBLIC-GOOD-RELATED ISSUES

(a) Regulations on work-related issues



(b) Regulations on access to local public goods



This figure shows the most frequent words of the migrant-related regulations on work topics (1,567 out of 4,273) and local public good access topics (286 out of 4,273). A larger font indicates a higher frequency. Appendix A.3 shows the corresponding word clouds for other topics: family planning policy, Hukou, social insurance, and miscellaneous issues.

example, some regulations are about enforcing the timely payment of wages to migrant workers before the Chinese New Year in a particular year. Although the enforcement is likely to improve migrant well-being, the effect might be one-off and short-lived. In rare cases where the enforcement is very detailed, I code it as 2 (−2 if it is against migrants). (3) If the regulation addresses *two or more issues*, either about guidelines or about enforcement, a score of 2 is assigned (−2 if it is against migrants).

Finally, after coding each regulation, I sum the scores of all regulations enacted up to a certain year to measure overall

migrant-friendliness, since each additional regulation either addresses a different issue or reinforces (or mitigates) the changes in the same issue. For example, Beijing's migrant friendliness in 2007 is the sum of the scores of regulations enacted from 1978 to 2007. The rationale is that local governments maintain a uniform level of migrant-friendliness since 1978 unless they state otherwise (by enacting a new regulation). Let us consider a prefecture that enacted a regulation that allows migrant children to attend local primary schools in year t and a regulation that provides vocational training for migrant workers in year $t + 1$. Both regulations are promigrant and address different issues on how to improve migrant well-being. Thus not only the average

regulation score but also the number of regulations is important for measuring migrant friendliness of the prefecture. Overall, migrant amenities should be an increasing function of the regulation score.

B. *Alternative Coding and Machine Learning Methods*

To ensure the objectivity of the score coding, I hired two research assistants with law degrees to code the scores independently. This was done at the early stage of the study through the online platform www.upwork.com, and the two research assistants coded 686 regulations enacted between 1994 and 2006 by provincial government and prefecture-level government. The correlations between my coding and their codings for individual regulation scores are 0.62 and 0.72 for the entire set of 686 regulations, and are 0.67 and 0.74 for the 390 regulations enacted by prefecture-level governments.

With this set of triple-coded regulations, I implement a supervised learning exercise with three different natural language processing (NLP) methods to supplement the manual coding. In each method, the algorithm processes the regulation text in a training dataset, builds a model, and calibrates the model parameters to maximize the probability of correctly predicting the outcome (the regulation score). I use the 686 triple-coded regulations as the training set and take the mean of the three scores, rounded to the closest integer, as the “true” regulation score for the training set. The rounding is to create a discrete set of outcomes, which is suitable for the classification exercise in NLP. Both the prefecture-level regulations and the province-level regulations are included to obtain a larger sample size. Then the model is applied to all 4,273 regulations to generate a score in the set of $\{-2, -1, 0, 1, 2\}$. Appendix table A1 presents the relevant number of regulations in different stages of the study.

The first NLP classification method is random forest (RF), where the algorithm is a decision tree combined with bootstrapped samples. The regulation text is decomposed into words, and each regulation is represented as a combination of words and corresponding frequencies of the words appearing in the text. The decision tree starts by taking an attribute (a word) and the value of the attribute (the word count), and splits the set of regulations by a criteria (word count larger or smaller than a number). Then the tree grows by moving on to the next attribute. Eventually the set of regulations are classified into groups taking the value in the set of $\{-2, -1, 0, 1, 2\}$. The order of attributes and their values used in the hierarchy of decision making are chosen to minimize the error rate, which is the share of misclassified regulations according to the rule. The final model is constructed using 686 regulations and 10,243 words, and the random forest algorithm correctly classifies 684 out of 686 regulations in the training set.

Once the trees are grown, the importance of the each attribute can be calculated, so one can see what the key words are in the decision making process. The top 200 words by

importance are plotted in appendix figure A2. The top three are “floating population,” “migrant worker,” and “temporary residence permit.” Among the rest, “family planning,” “obtain local Hukou,” and “(wage) arrear” are important words. “Timeliness,” “effectiveness,” and “normativity” are words usually used in the regulations to present a positive sentiment, and “rental housing” and “public security” are used in the ones coded more negatively.

After building the random forest model, I use the model to code all 4,273 regulations. The random forest coding is highly correlated with the author’s manual coding, with a correlation of 0.73 among the 4,273 regulations. I also use two other neural-network-based deep learning methods for robustness: multilayer perceptron classification (MLP) and convolutional neural network (CNN) with word-embedding. The main advantage of the random forest method is its interpretability, since it uses words and word frequencies to make the classification with a decision tree, and one can see clearly the importance of words used in the decision tree to classify regulations into categories. In contrast, the MLP and CNN methods are based on neural networks and use different information and algorithms, whose results are more difficult to interpret than those of the random forest. MLP and CNN methods generate very similar coding results, and in the empirical analysis, all three machine-learning codings will be used, with very similar results.¹⁵

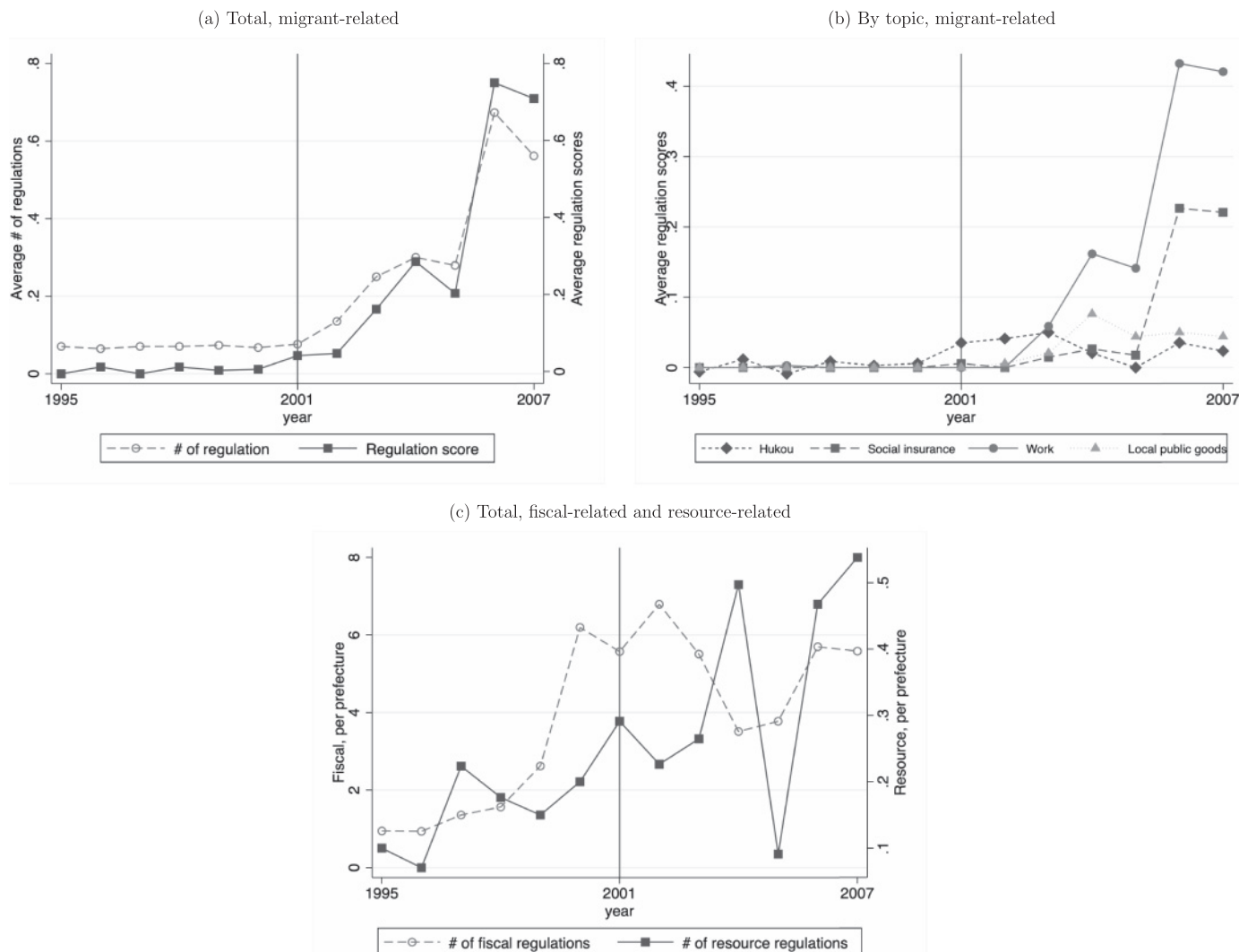
C. *Summary of Statistics of the Regulation Data*

Over time, the number of migrant-related regulations and their migrant friendliness increased. Figure 2(a) plots the prefecture-level average number of new regulations on migrant issues (dashed line) and the corresponding average regulation scores (solid line). The trend shows that before 2001, about 0.07 new regulation per prefecture per year addressed migrant issues. However, the migrant-friendliness score was essentially zero, indicating no deviation from the national policy, on average. After 2001, both the number of new regulations and the regulation friendliness score increased substantially. In 2006, for example, there was 0.67 new regulation per prefecture with an average score of 0.75, indicating that there were more regulations and those regulations were more favorable to migrants than before 2001. Panel b plots the trends in regulation score by topic. The increase in the total score of regulation is mainly driven by work- and social-insurance-related regulations (the solid line with circles and dashed line with squares, respectively). In 2006, the average score 0.43 is for work-related regulations and 0.23 is for social-insurance-related ones.

One potential concern about the regulation data is that the number of migrant-related regulations might increase mechanically due to improved data coverage over time. To alleviate this concern, I also count the total number of regulations on fiscal topics and resource topics. These regulations

¹⁵The details of the data and the coding methods in appendix A.

FIGURE 2.—THE NUMBER OF NEWLY-ENACTED MIGRANT-RELATED REGULATIONS AND REGULATION SCORES (1995–2007), TOTAL AND BY TOPIC (PANEL A AND PANEL B), AND THE NUMBER OF REGULATIONS IN THE FISCAL-RELATED AND RESOURCE-RELATED CATEGORIES (1995–2007, PANEL C)



Each dot is the average of 340 prefectures in a particular year. Panel a presents the trend in the prefecture-level average number of newly enacted migrant-related regulations (dashed line, vertical axis on the left) and the prefecture-level average score of these regulations (solid line, vertical axis on the right). Panel b presents the trends in the prefecture-level average score of newly-enacted migrant-related regulations by topic. In panel c, the dashed line represents the prefecture-level average number of fiscal regulations (vertical axis on the left). The solid line represents the prefecture-level average number of resource-related regulations (vertical axis on the right). The vertical line corresponds to China's accession to the WTO in 2001.

are more routine and logistic-related and should be less responsive to economic conditions. Figure 2(c) plots the fiscal regulations (the dashed line) and the resource regulation (the solid line). Both of them act as a kind of placebo regulation; neither line demonstrates a clear pattern, and there is no trend break around 2001, in contrast to the migration related regulations in figures 2(a) and 2(b).

IV. China's WTO Accession and Declines in Trade Barriers

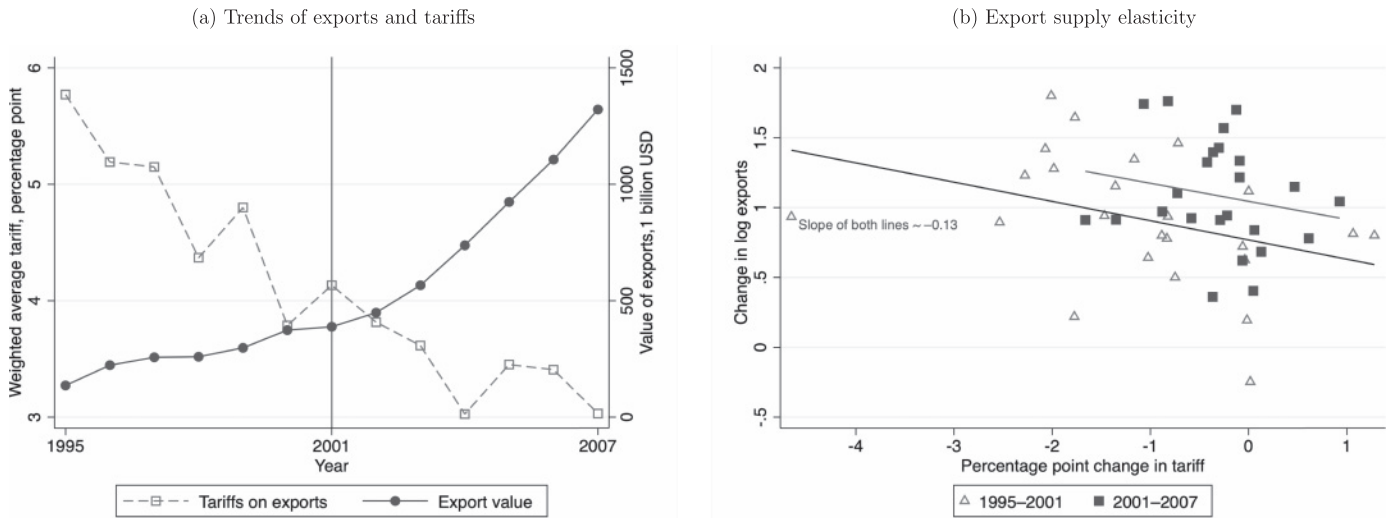
A. Prefecture-Level Tariff Changes

China entered the WTO in November 2001 as the 143rd member country. In the accession agreement, China and the partner countries committed to reducing import tar-

iffs, removing quotas, and reducing other nontariff barriers. In short, China started to enjoy MFN status. This means, among other things, that Chinese goods would face the same tariffs as other WTO members.

This paper focuses on the decline in output tariffs on Chinese goods imposed by countries that import from China. The 2-digit-SIC-level tariff is constructed as the weighted average of destination-country applied tariffs on Chinese exports in the specific two-digit industry, where the weights are shares of exports in this destination country in the specific industry in 1995, using the World Bank TRAINS dataset. The decline was sizable, and industries that experienced bigger tariff declines also experienced bigger export volume growth. Figure 3 panel a shows that the tariff on Chinese goods stood at about 5.8 percentage points in 1995, declined to 4.1 percentage points in 2001, and declined even further

FIGURE 3.—THE TRENDS OF TARIFFS ON EXPORTS AND EXPORT VALUES (1995–2007) AND EXPORT SUPPLY ELASTICITIES (1995–2001 AND 2001–2007)



In Panel a, the dashed line presents trend of the weighted average of industrial-level tariffs on exports, where the weights are shares of exports in this industry. The industry-level tariff on exports is constructed as the weighted average of destination-country tariffs on Chinese exports in the specific industry, where the weights are shares of exports in this destination country in the specific industry in 1995. The solid line presents the trend in the value of exports. Panel b presents the relationship between percentage point change in tariff and change in log export by industry, where the triangles are changes from 1995 to 2001 and squares are changes from 2001 to 2007.

to 3 percentage points in 2007. Figure 3(b) shows percentage point changes in tariffs on the horizontal axis and the change in log exports on the vertical axis. Each dot represents an industry. The hollow triangles are for 1995–2001 and the solid squares are for 2001–2007. In both periods, the fitted lines have a slope of -0.13 , meaning that a one percentage-point reduction in the tariff faced by Chinese exporters induces a 13%–14% increase in export values.

I study the post-WTO period of 2001–2007, comparing prefectures that had bigger versus smaller declines in tariffs on exports. Although it seems that there are no discontinuous changes in the overall tariff level from the pre-WTO period to the post-WTO period, there are substantial changes at the industry level.¹⁶ My identification strategy will rely on differences in export specialization across Chinese prefectures, and as in Kovak (2013), the regional change in tariffs on exports in prefecture i and from time t to t' is

$$\Delta \text{tariffs on exports}_{it} = \sum_j \beta_{ij} [\ln(1 + \tau_{jt}^X) - \ln(1 + \tau_{jt'}^X)],$$

$$\text{where } \beta_{ij} = \frac{\lambda_{ij} \frac{1}{\theta_{ij}}}{\sum_{j'} \lambda_{ij'} \frac{1}{\theta_{ij'}}},$$

$\lambda_{ij} = \frac{L_{ij}}{L_i}$ is the fraction of regional labor allocated to industry j , and $1 - \theta_{ij}$ is the cost share of labor in industry j . λ_{ij} and θ_{ij} are calculated from the 2000 Industrial Enterprises Survey data. τ_{jt}^X is the tariff on exports in industry j in time

t , and it is measured using average tariffs faced by Chinese exporters.¹⁷

I focus on the changes in tariffs on exports for several reasons. First, the decline in tariffs on exports was direct and salient from the government officials' perspective. The decline in tariffs on inputs also played an important role, but it was more indirect. Second, although policy discussions mentioned import competition, they mostly focused on the competition on high-end goods such as automobiles and agricultural products. China's comparative advantage was thought to be on labor-intensive or low-skill-intensive goods, and the export expansion in these industries was likely to trigger migration regulation changes. In the empirical analysis, I control for changes in tariffs on imports and changes in tariffs on intermediate inputs.

B. Trade Uncertainty and Quotas

The WTO accession also induced other positive demand shocks for Chinese exports. First, Pierce and Schott (2016) and Handley and Limão (2017) show that the United States applied MFN tariffs on Chinese exports even before the WTO accession. However, before 2001, there was great uncertainty regarding the U.S. trade policy: the MFN status had to be approved each year by the Senate and the House; otherwise, the Column 2 tariff would be applied to Chinese exports. Handley and Limão (2017) argue that the greater policy certainty was the main impact of the WTO accession on the U.S.–China trade relationship. I measure the size of the

¹⁶Appendix figure B1 shows the distribution of tariff reductions across industries.

¹⁷The regional changes in tariffs on imports and intermediate inputs are measured in a similar way. See appendix B.1 for details.

PNTR shock experienced by prefecture i in the time period $t = 2001-2007$ as

$$PNTR_{it} = \sum_j \beta_{ij} (Column2_{j,2000}^{US} - MFN_{j,2000}^{US}) \times \frac{export_{i,2000}^{US}}{export_{i,2000}^W},$$

where $Column2_{j,2000}^{US}$ is the U.S. Column 2 tariff on industry j in 2000, $MFN_{j,2000}^{US}$ is the U.S. MFN tariff, $export_{i,2000}^{US}$ is the total exports from prefecture i to the United States in 2000, and $export_{i,2000}^W$ is the total exports from China in 2000.¹⁸

Second, Khandelwal et al. (2013) show that the Chinese textile and clothing exports to the United States, the European Union, and Canada were subject to Multi Fibre Arrangement (MFA) quota restrictions until January 2005. The removal of these restrictions boosted Chinese exports in corresponding industries. A prefecture i 's exposure to MFA shock in the time period $t = 2001 - 2007$ is

$$MFA_{it} = \sum_p \frac{export_{p,i,2000}}{\sum_{p'} export_{p',i,2000}} \cdot D(MFA_p^{2001-2005} = 1),$$

where p is an eight-digit HS category, $export_{p,i,2000}$ is the export of product p from Chinese prefecture i to the world, and $D(MFA_p^{2001-2005} = 1)$ is an indicator variable that takes the value of 1 if the export is to the United States, Canada, and the European Union, and product p is subject to the MFA quota at any time between 2001 and 2005.

Seeing the opportunities arising from these shocks to promote exports, Chinese local governments may respond similarly by relaxing internal migration restrictions. I will investigate the impact of these export demand shocks on migration regulations empirically.

V. Prefecture Sample Restriction

Mainland China is composed of 31 provinces, which are divided into 340 prefectures, and each prefecture contains rural areas and urban areas. I restrict the sample of prefectures used in different empirical specifications.

First, in the main empirical analysis, I always control for the changes in tariffs on imports and intermediate inputs. Thus I drop all seven prefectures in the Province of Tibet due to their missing input-output matrix, resulting in 333 prefectures.

Second, the petroleum industry is distributed very unevenly across prefectures: 293 of the 333 prefectures have zero employment in petroleum industry, and the 95th percentile of the distribution is 5.5%. As shown in appendix figure B2, the Herfindahl index of the petroleum industry is the

highest. In addition, the petroleum industry is entirely state-owned. Thus I drop 16 prefectures with petroleum industry employment shares above 5.5%, since they are likely to be very different from the rest of the prefectures, resulting in 317 prefectures.

Third, in the empirical analysis, I will investigate whether prefectures with different income levels responded to the changes in tariffs on exports differentially, so I use the per capita GDP and wage data from the *Prefecture Statistic Yearbook*. However, the Yearbook only covers relatively urban prefectures. As shown in appendix figure B3, the median employment share in the agriculture sector is 66% in prefectures that are included in the 2001 Yearbook, and 82% in prefectures that are not included.¹⁹ Thus, in most of regressions, I use the 248 prefectures that are included in the 2001 *Prefecture Statistics Yearbook* as the main sample.

Finally, to investigate whether prefectures with different changes in tariffs on exports after 2001 experienced differential wage and GDP growth in the pre-period (1995–2001), I use 235 prefectures whose wage and GDP data are included in the 1995 *Prefecture Statistics Yearbook*.

VI. Main Empirical Results: Effects of Changes in Tariffs on Exports on Regulation Changes

A. Econometric Framework and Identification

Did trade induce changes in labor regulations? I investigate this question using the following equation:

$$\Delta \sinh^{-1}(\text{regulation score})_{it} = \alpha_0 + \alpha_1 \Delta \text{tariffs on exports}_{it} + X_{it} \Gamma + \epsilon_{it},$$

where i represents a prefecture, t represents a time period, Δ represents the change during the time period. The main explanatory variable is the change in tariffs on exports, and the outcome, $\Delta \sinh^{-1}(\text{regulation score})_{it}$ is the change in the inverse-hyperbolic-sine transformation of the regulation score in prefecture i and time period t . I use the inverse-hyperbolic-sine-transformed regulation score instead of the levels of the regulation score because the distribution of levels is very skewed to the right: the mean of the change in regulation score from 2001 to 2007 is 2.2, while the maximum is 37. The corresponding numbers for the transformed ones are 0.86 and 4.57, respectively. The inverse hyperbolic sine transformation allows me to include the prefecture years with zero or negative cumulative migrant friendliness. X_{it} is a matrix of potential confounding factors that could be correlated with the changes in tariffs on exports.

The parameter α_1 is the coefficient of the effect of changes in tariffs on exports on changes in migration regulation score. A negative α_1 indicates that a larger decline in tariffs

¹⁸The 2000 Column 2 tariffs and MFN tariffs at the eight-digit HS level imposed by the United States is from Handley and Limão (2017). I then use Pierce and Schott's (2012) concordance to map the eight-digit HS tariffs to two-digit SIC tariffs. Each prefecture's exports to the U.S. and other countries are from the 2000 customs data with destination country and source region information.

¹⁹A similar urban-rural divide shows up when I compare them by the share of urban population. The employment share in agriculture and the urban population share are from the 2000 population census, which includes all 340 prefectures.

on exports will induce growth in exports and labor demand, resulting in an increase in migrant friendliness. Identification of α_1 requires that conditional on X_{it} , there are no unobservables that are correlated with the changes in tariffs on exports and have a direct impact on migration regulation changes. I address two types of identification issues here. First, at the industry level, the WTO-induced decline in tariffs on Chinese exports should be uncorrelated with pre-WTO trends, such as pre-WTO export growth and pre-WTO tariff reductions. Otherwise, the regional changes in tariffs on exports will capture preexisting industry characteristics instead of WTO-induced changes. Second, at the prefecture level, the WTO-induced regional changes in tariffs on exports should not be correlated with pre-WTO trends, such as pre-WTO GDP growth and wage growth.

China obtained MFN status after the WTO accession, and the resulting tariff reductions on Chinese exports were mainly determined by WTO rules. Empirically, across industries, the post-WTO tariff declines could not be predicted by either the pre-WTO export growth or the pre-WTO tariff decline. Thus there were still relative industry “winners” and “losers” due to the WTO accession. Figure 4(a) plots the percentage-point change in tariffs on exports in the 2001–2007 period on the y axis and the percentage change in exports in the 1995–2001 period on the x axis. The linear fitted line has an insignificant coefficient of 0.03, meaning that the industries that had bigger pre-WTO export growth were not the ones that experienced bigger post-WTO tariff cuts. Figure 4(b) plots the percentage-point change in tariffs on exports from 2001 to 2007 against the percentage-point change in tariffs on exports from 1995 to 2001. The linear fitted line has a coefficient of -0.03 and is statistically insignificant. This indicates that the industries experiencing larger tariff declines after the WTO accession had similar export growth and tariff changes in prior years.

Declining tariffs in various industries translated into prefecture-level tariff changes that should not be correlated with local economic conditions other than through the prefecture-level industrial composition. Figure 4(c) plots the trends in wages of prefectures with small (bottom third of the decline in tariffs on exports distribution in the 2001–2007 period), medium (middle third), and large (top third) declines in tariffs on exports, and the three trends from 1995 to 2001 are not statistically different from each other. Figure 4(d) plots the trends in per capita GDP; here, there seems to be a slight divergence among the three groups from 1995–2001. The three trends are not statistically different from each other, but to be conservative, I control for 1995–2001 wage and per capita GDP growth in the regressions.

B. Main Results

Figure 5 shows the relationship between prefecture-level changes in tariffs on exports and changes in migration regulation scores from 2001 to 2007. The horizontal axis depicts the change in tariffs on exports in the 2001–2007 pe-

riod, and the vertical axis depicts the post-WTO change in the inverse-hyperbolic-sine-transformed regulation score. The dashed line is the linear fitted line with 2001 population size as weights, and the dotted line is the unweighted linear fitted line. The slope ranges from -1.8 to -0.9 , and the values are statistically significant at the 5% level. This indicates that prefectures that experienced larger declines in tariffs on exports saw their regulation score rise, meaning that they became more friendly to migrants. By comparison, the same regressions in the pre-WTO period give slopes of -0.01 and 0.04 , and they are statistically insignificant.²⁰ In addition, following Borusyak et al. (2018), I aggregate the regulation changes and changes in tariffs on exports to the industry level to have a more transparent demonstration of the industry-level variation in appendix C.3.²¹

Table 1, panel A presents the corresponding regression results using the manually coded regulation score. The outcome variable is the change in the inverse-hyperbolic-sine-transformed regulation score from 2001 to 2007. All columns have standard errors clustered at the province level to account for potential spatial correlation of laws and regulations at the province level. Column 1 includes 333 prefectures and controls for changes in tariffs on exports, imports, inputs from 2001 to 2007, and also the initial regulation score in 2001. The coefficient -1.04 is statistically significant at the 1% level. It implies that a 1-percentage-point larger decline in tariffs on exports increased the change in the inverse-hyperbolic-sine-transformed regulation score by 1.04, which is equivalent to a one-standard-deviation larger regulation score increase. I divide prefectures into three groups: (1) prefectures with large decline in tariffs on exports (0.36-percentage-point tariff decline on average); (2) medium decline (0.17 percentage point); and (3) small decline (-0.04 percentage point). Thus, compared with small-decline prefectures, the big-decline prefectures experienced a 0.42-unit larger increase in the inverse-hyperbolic-sine-transformed regulation score; the difference is equal to 0.40 standard deviation of the score increase in the 2001–2007 period.

Column 2 controls for the changes in tariffs on exports, imports, and inputs from 1995 to 2001 and also adds the MFA shock and the measure of uncertainty reduction due to the PNTR shock. The coefficient estimate for the uncertainty shock is 4.67 and statistically significant, indicating that a one-standard-deviation larger uncertainty shock (0.03) led to a 0.14 unit larger increase in the inverse-hyperbolic-sine-transformed regulation score. The coefficient estimate for the MFA shock is 5.43 and statistically significant, with a one-standard-deviation larger MFA shock (0.01) leading

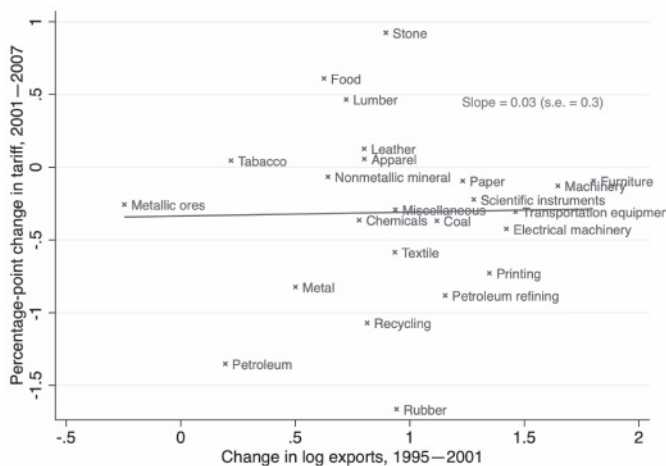
²⁰See appendix C.2 for the corresponding pre-WTO plot.

²¹Both Borusyak et al. (2018) and Goldsmith-Pinkham et al. (2020) propose diagnostic tests for exogeneity of the Bartik measure. However, both papers use Autor et al. (2013) as an example of application, where there are over 200 industries. Borusyak et al. (2018) points out that the application is less appropriate for settings with a small number of industries, as in Kovak (2013).

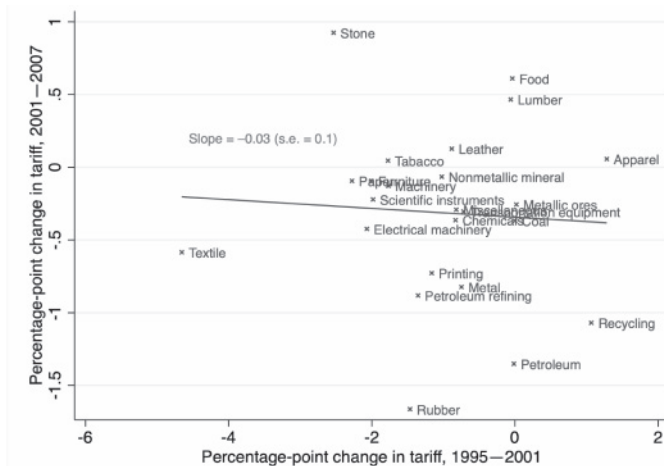
FIGURE 4.—PRE-TRENDS AT THE INDUSTRY LEVEL AND AT THE PREFECTURE LEVEL

A. The correlation between changes in tariffs on exports (2001–2007) with changes in exports values (1995–2001), and with changes in tariffs on exports (1995–2001)

(a) Pre-WTO exports

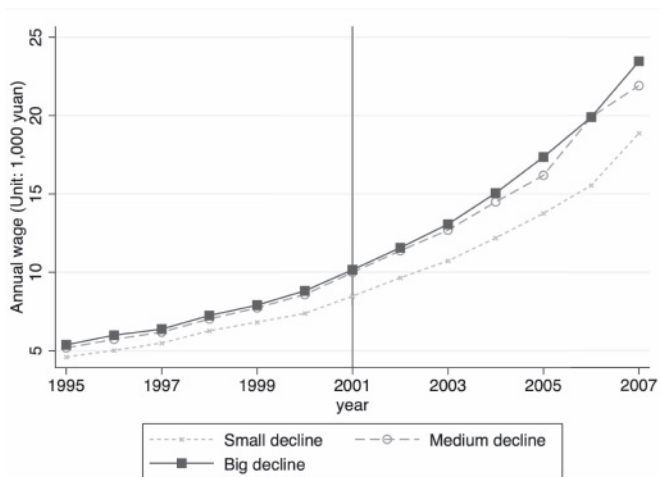


(b) Pre-WTO tariffs

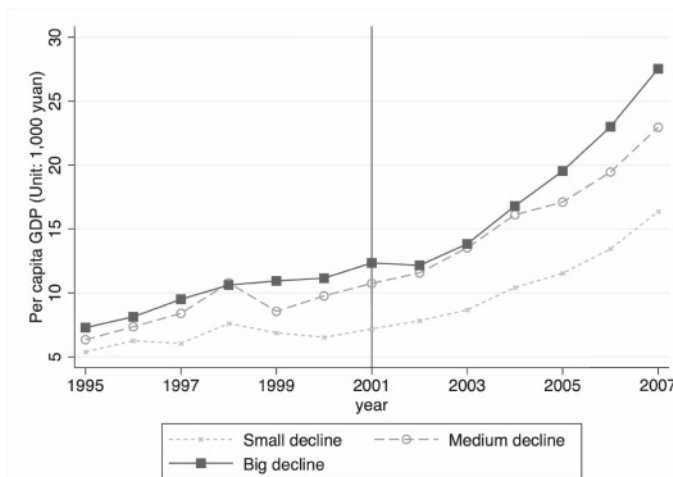


B. Prefecture-level trends in wages and per capita GDP (1995–2007) by size of declines in tariffs on exports (2001–2007), 235 prefectures

(c) Wages



(d) Per capita GDP



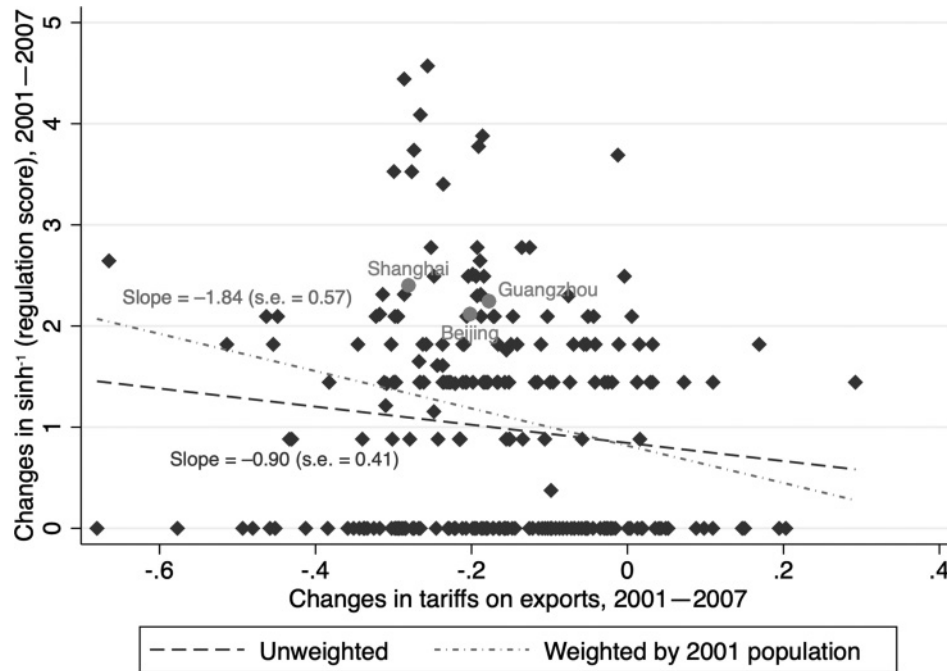
Panels a and b present the relationship between tariff changes in the post-WTO period and pre-WTO tariff and export changes at the industry level. Each dot is a two-digit SIC industry. Panel (a) shows the relationship between the change in log exports in the 1995–2001 period and the percentage-point change in tariffs on exports in the 2001–2007 period. Panel b shows the relationship between the percentage-point change in tariffs on exports in the 1995–2001 period and the percentage-point change in tariffs on exports in the 2001–2007 period. Panels c and d present the trends in wages and GDP for prefectures in different groups. Prefectures are divided into small-, medium-, and large-tariff-decline groups, where the small-decline regions are at the bottom third of the distribution of declines in tariffs on exports (2001–2007), the medium ones at the medium third, and the large ones at the top third. Panel c presents the trends in average wages by prefecture group, and panel d presents the trends in average per capita GDP by prefecture group. The sample includes 235 prefectures with GDP and wage data from the *Prefecture Statistics Yearbooks* in 1995, 2001, and 2007.

to a 0.05 unit larger increase in the inverse-hyperbolic-sine-transformed regulation score. The coefficients on the changes in tariffs on exports remain largely unchanged, and the effect of the change in tariffs on exports is larger than the other two shocks. That is—a one-standard-deviation larger decline in the tariffs on exports led to a 0.18 unit larger increase in the inverse-hyperbolic-sine-transformed regulation score. This result is reasonable since the tariff changes affected all industries and all regions, the PNTR shock af-

fected mostly the firms and regions that exported to the United States, and the MFA shock affected mostly the textile industry.

Columns 3–7 use different samples and control for various potential confounding factors to check the robustness of the result. Columns 3 and 4 replicate columns 1 and 2 but exclude the 16 prefectures with high employment shares in the petroleum industry. Columns 5 and 6 further exclude 69 prefectures with missing GDP and wage measures in 2001.

FIGURE 5.—THE RELATIONSHIP BETWEEN PREFECTURE-LEVEL CHANGES IN TARIFFS ON EXPORTS AND REGULATION CHANGES (2001–2007), 248 PREFECTURES



Each dot is a prefecture. This figure presents the relationship between changes in tariffs on exports and changes in the inverse-hyperbolic-sine-transformed regulation score from 2001 to 2007. The dashed line is the linear fitted line, unweighted, and the dotted line is the linear fitted line weighted by the 2001 population size. The sample includes 248 prefectures with GDP, wage, and population data from the 2001 *Prefecture Statistics Yearbook*.

Column 7 further controls for other variables that might have been important determinants of regulation changes, including lagged wage and GDP growth. The estimates for the effect of changes in tariffs on exports remain stable and are all statistically significant at the 10% level; all of them are within the 95% confidence interval of the estimate in column 1.

In all regressions, the standard errors are clustered at the province level. Given the small number of provinces, I also calculate bootstrapped standard errors. I then consider the estimates suggested by Adao et al. (2019) and report the p values using the $AKM(0)$ and $AKM(1)$ estimates. The p values of the coefficient estimate for the effect of changes in tariffs on exports are similar to ones in the main specification.

Panel B repeats the exercise with regulation scores coded using the random forest method. The results are very similar to the ones in panel A; as shown in appendix figure C1, the correlation between scores coded manually and coded using the random forest method is close to one. Appendix table C1 shows that results are similar when I use MLP and CNN measures separately and when I use the average of the three NLP scores.

Overall, the findings are consistent with the hypothesis that in the post-WTO period, prefectures with larger declines in tariffs on exports relaxed migration restrictions more. Appendix C.5 shows a stacked long-difference specification where each prefecture has two observations (1995–

2001 and 2001–2007), and the results are similar. The effects are also robust to using the levels of regulation scores instead of the inverse-hyperbolic-sine-transformed ones (appendix C.6). Appendix C.7 decomposes the migration regulations by topic and finds that the work-related and local-public-good-related regulations were impacted the most by changes in tariffs on exports. The results are robust to using a three-point coding scale of regulations instead of a five-point coding scale. In appendix C.8, I construct alternative Bartik-style trade exposures following Autor et al. (2013) and instrument the Chinese export growth with importing countries' income growth; the results show similar patterns. Finally, to address the concern that certain industries drive the regional tariff variation and are correlated with other local factors that affect regulation changes directly, in appendix C.9, I add industry employment shares one at a time and find that the result is not sensitive to specific-industry effects.

A prefecture may not only respond to its own tariff changes but also tariff changes happening in other prefectures, since all prefectures are competing for the migrant labor supply. In addition, a prefecture may change its own migrant regulations when other prefectures change theirs. In appendix C.10, I show that prefectures responded strongly to changes in tariffs on exports and regulation changes in other prefectures with similar income levels. This is consistent with the fact that Chinese prefectures in different income groups are considered to be in different tiers (e.g.,

TABLE 1.—EFFECTS OF CHANGES IN TARIFFS ON EXPORTS ON CHANGES IN REGULATION SCORES (2001–2007), MANUAL CODING AND RANDOM FOREST CODING

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A.	Dependent variable: $\Delta \sinh^{-1}$ (regulation score), 2001–2007, manual coding						
Δ tariffs on exports, 2001–2007	–1.04*** (0.30)	–0.91*** (0.26)	–1.48*** (0.33)	–1.20*** (0.32)	–1.60*** (0.50)	–1.38*** (0.49)	–1.02* (0.53)
Δ tariffs on imports, 2001–2007	–0.03 (0.04)	–0.01 (0.04)	–0.05 (0.03)	–0.04 (0.04)	–0.11** (0.04)	–0.08 (0.07)	–0.08 (0.07)
Δ tariffs on inputs, 2001–2007	–0.54*** (0.14)	–0.16 (0.20)	–0.53*** (0.14)	–0.21 (0.20)	–0.31** (0.14)	–0.04 (0.23)	–0.13 (0.23)
$\sinh^{-1}$ (regulation score), 2001	–0.09 (0.24)	–0.11 (0.22)	–0.10 (0.23)	–0.11 (0.22)	–0.12 (0.22)	–0.12 (0.21)	–0.15 (0.20)
PNTR shock, 2001–2007		4.67*** (1.44)		3.99*** (1.25)		2.89** (1.27)	1.03 (1.21)
MFA shock, 2001–2007		5.43*** (1.94)		4.89** (1.91)		4.20** (1.91)	4.86** (2.25)
Δ tariffs on exports, 1995–2001		–0.05 (0.16)		–0.04 (0.17)		–0.12 (0.21)	–0.17 (0.23)
Δ tariffs on imports, 1995–2001		–0.00 (0.02)		0.00 (0.02)		0.02 (0.03)	0.02 (0.02)
Δ tariffs on inputs, 1995–2001		–0.22** (0.10)		–0.20** (0.09)		–0.25** (0.12)	–0.14 (0.13)
Δ log wage, 1995–2001							0.69 (0.52)
Δ log GDP p.c., 1995–2001							0.65*** (0.22)
Observations	333	333	317	317	248	248	235
R-squared	0.09	0.14	0.11	0.15	0.08	0.11	0.15
Bootstrap p-value, Δ tariffs on exports	.006	.005	0	.003	.006	.014	.078
AKM1 p-value, Δ tariffs on exports	.004	0	.001	.002	.015	.022	.056
AKM0 p-value, Δ tariffs on exports	.024	.02	.02	.036	.036	.055	.051
Panel B.	Dependent variable: $\Delta \sinh^{-1}$ (regulation score), 2001–2007, random forest coding						
Δ tariffs on exports, 2001–2007	–1.00*** (0.30)	–0.86*** (0.26)	–1.39*** (0.33)	–1.06*** (0.30)	–1.46*** (0.49)	–1.21** (0.47)	–0.89* (0.52)
Observations	333	333	317	317	248	248	235
R-squared	0.09	0.16	0.11	0.16	0.07	0.12	0.16
Mean (sd) $\Delta \sinh^{-1}$ (regulation score), manual	0.86 (1.04)		0.86 (1.04)		1.00 (1.08)		1.03 (1.08)
Mean (sd) $\Delta \sinh^{-1}$ (regulation score), RF	0.87 (1.04)		0.87 (1.03)		1.01 (1.06)		1.04 (1.07)
Mean (sd) Δ tariffs on exports	–0.16 (0.20)		–0.14 (0.18)		–0.18 (0.15)		–0.18 (0.15)

In panel A, the dependent variable is the change in the inverse-hyperbolic-sine-transformed regulation score using manual coding. In panel B, the dependent variable is the change in the inverse-hyperbolic-sine-transformed regulation score using the random forest coding, and the controls are the same as in Panel A. Bootstrap p values are calculated using the wild bootstrap method with 1,000 repetitions, and AKM1 and AKM0 p values are calculated using the method in Adao et al. (2019). In columns 1 and 2, the sample include 333 prefectures, where 7 prefectures in Tibet are excluded since there is no input-output table. In columns 3 and 4, the sample exclude 16 prefectures with a top 5% share of employment in petroleum industry (larger than 0.055). In columns 5 and 6, the sample excludes 69 prefectures that are not included in the 2001–2007 *Prefecture Statistics Yearbook* (and are without the measure of GDP and wage). In column 7, the sample excludes 13 prefectures without measures of GDP and wage changes from 1995 to 2001. Standard errors are clustered at the province level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Beijing, Shanghai, and Guangzhou are in the first tier), and they are more likely to compete within tiers for capital, labor, and other resources.²²

C. Heterogeneous Effects

Prefectures with more migrant-intensive industries should respond more strongly to the changes in tariffs on exports. I investigate this heterogeneous effect using three sets of empirical proxies for a prefecture's migrant intensity. First, I calculate the migrant share of employment in each industry using the 2005 mini-census. Then the regional migrant in-

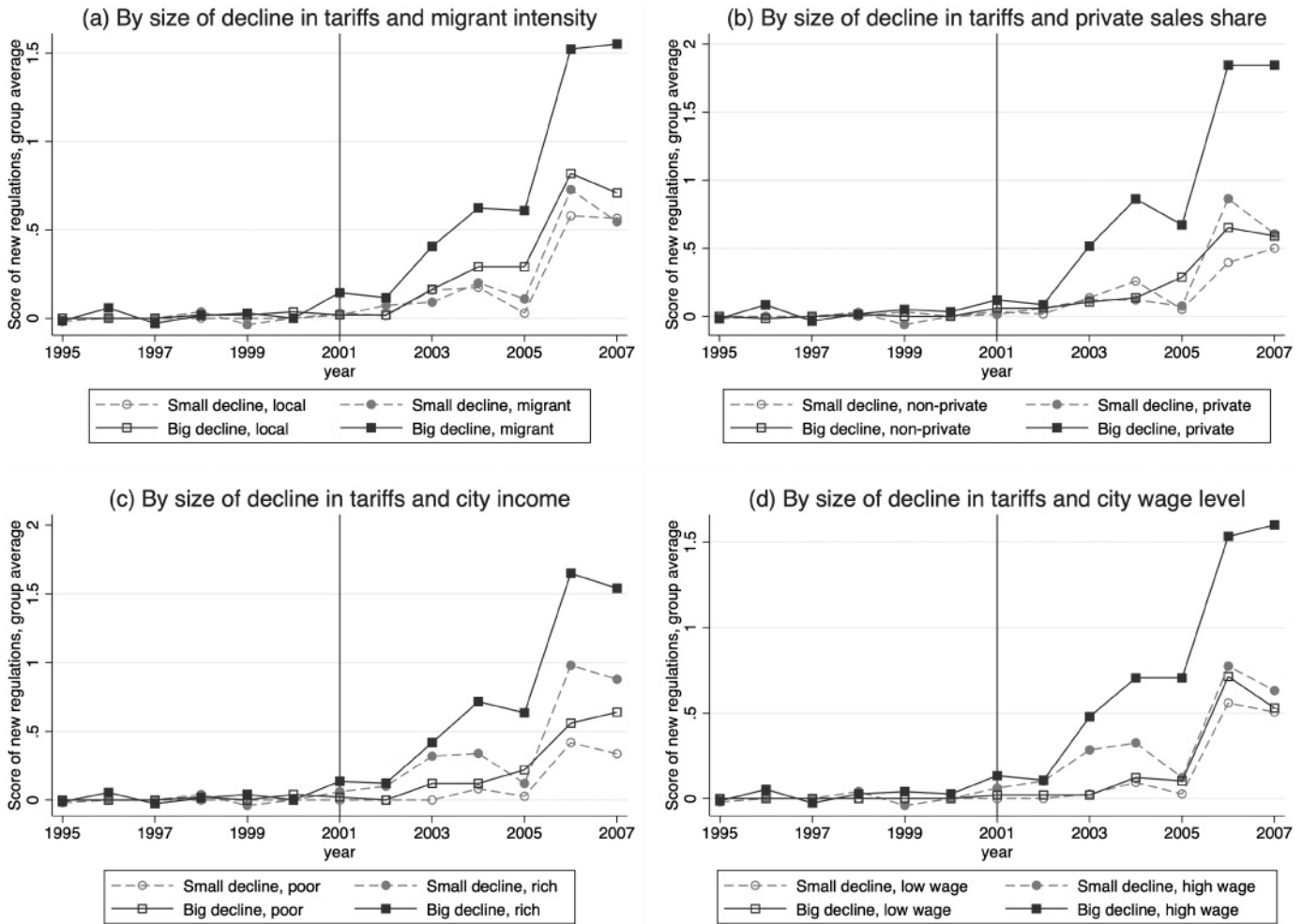
tensity is the employment-weighted average migrant share of employment across industries. Second, state-owned enterprises are usually more restrictive in Hukou requirements and hire more locals than migrants. Thus the share of total sales by private firms will be positively correlated with migrant intensity.²³ Third, a prefecture's income level is empirically positively correlated with migrant intensity. This could be because richer prefectures tend to have more diversified industrial composition and rely less on state-owned enterprises. Thus per capita GDP and wages can act as proxies for the migrant intensity.

Figure 6 divides prefectures into four groups, depending on the 2001–2007 change in tariffs on exports and one of the four proxies for migrant intensity in 2001, with the

²²This result is in line with Tiebout (1956) type of fiscal policy decisions. Feler and Henderson (2011) find negative strategic interactions among localities in making exclusionary policies against low-income households: Brazilian cities seem to view low-income households mainly as causes of fiscal expenditure instead of sources of fiscal revenue.

²³I use the 2001 Industrial Enterprises Survey data to calculate the prefecture-level share of total sales in private enterprises.

FIGURE 6.—AVERAGE PREFECTURE-LEVEL REGULATION CHANGES (1995–2007), BY THE SIZE OF THE CHANGE IN TARIFFS ON EXPORTS (2001–2007) AND BY MIGRANT INTENSITY (2001), 248 PREFECTURES



Panel a divides prefectures into four groups. The small-decline and local group represents prefectures whose post-WTO decline in tariffs on exports was below the median and migrant intensity was below the median. Panel b uses the 2001 prefecture-level sales share of private firms as the measure for migrant intensity; panel c uses the 2001 prefecture-level log per capita GDP; and panel d uses the 2001 prefecture-level wage. The sample includes 248 prefectures with GDP, wage, and population data from the 2001 *Prefecture Statistics Yearbook*.

median value as the cutoff. In panel a, the four groups are (1) prefectures with a large decline in tariffs on exports and migrant-intensive (solid line with solid squares), (2) prefectures with a large decline in tariffs on exports and non-migrant-intensive (solid line with hollow squares), (3) prefectures with a small decline in tariffs on exports and migrant-intensive (dashed line with solid dots), and (4) prefectures with a small decline in tariffs on exports and non-migrant-intensive (dashed line with hollow dots). In panels b, c, and d, the proxies for migrant intensity are the private firm sales share, per capita GDP, and wages, respectively.

The four graphs confirm the heterogeneous response to changes in tariffs on exports: prefectures that experienced larger declines in tariffs on exports and were more migrant-intensive had the largest change in migrant regulations, and prefectures that experienced smaller declines in tariffs on exports and were less migrant-intensive had the smallest change.

Table 2 shows similar findings as in figure 6 using regression analysis. The regression equation is as follows:

$$\Delta \sinh^{-1}(\text{regulation score})_{it} = \beta_0 + \beta_1 \Delta T_{it} + \beta_2 I_{it} + \beta_3 I_{it} * \Delta T_{it} + X_{it} \Gamma + \epsilon_{it},$$

where ΔT_{it} is the change in tariffs on exports in prefecture i and time period starting at $t = 2001$, I_{it} is one of the four measures for migrant intensity in prefecture i and year $t = 2001$. Table 2, Columns 1, 3, 5, and 7 replicate table 1, column 5, and additionally control for I and the interaction of changes in tariffs on exports with I . Columns 2, 4, 6, and 8 add additional controls such as pre-WTO tariff changes, PNTR shock and MFA shock, and pre-WTO wages and GDP growth, as in table 1 column 7.

Column 1 shows a negative interaction effect for migrant intensity and changes in tariffs on exports (-17.31), and a positive coefficient for changes in tariffs on exports (4.65).

TABLE 2.—HETEROGENEOUS EFFECTS OF CHANGES IN TARIFFS ON EXPORTS ON CHANGES IN REGULATIONS (2001–2007)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: $\Delta \sinh^{-1}(\text{regulation score})$, 2001–2007, manual coding							
Baseline characteristic:	Migrant intensity		Private firm share		Log(GDP p.c.)		Log(wage)	
Δ tariffs on exports 2001–2007	4.65 (3.65)	6.63 (3.90)	1.21 (0.72)	2.10** (0.91)	10.98* (5.97)	12.80** (5.86)	29.53 (18.72)	37.71** (18.26)
Baseline characteristic 2001	−0.87 (2.22)	−2.94 (2.49)	−0.50 (0.34)	−1.05** (0.44)	0.49*** (0.18)	0.48** (0.20)	0.54 (0.45)	0.46 (0.51)
Δ tariffs on exports $\times$ baseline chara.	−17.31 (10.93)	−21.89* (11.46)	−5.47*** (1.09)	−6.14*** (1.20)	−1.29* (0.68)	−1.47** (0.66)	−3.33 (2.07)	−4.18** (2.01)
Additional controls	X							
Observations	248	235	248	235	248	235	248	235
R-squared	0.09	0.16	0.13	0.20	0.23	0.27	0.15	0.21
Mean of baseline chara.	.34	.34	.55	.55	8.93	8.96	9.11	9.12
Sd. of baseline chara.	.05	.05	.23	.23	.64	.65	.28	.28

Columns 1, 3, 5, and 7 replicate table 1 column 5 and add a prefecture characteristic in 2001 and its interaction with the changes in tariffs on exports from 2001 to 2007. Columns 2, 4, 6, and 8 replicate table 1, column 7 and add a prefecture characteristic in 2001 and its interaction with the changes in tariffs on exports from 2001 to 2007. The baseline prefecture characteristic is the migrant intensity in 2001 in columns 1 and 2, the sales share of private firms in 2001 in columns 3 and 4, the log per capita GDP in 2001 in columns 5 and 6, and the log wage in 2001 in columns 7 and 8. The mean (s.d.) of $\Delta \sinh^{-1}(\text{regulation score})$ is 1.0 (1.1), and the mean (s.d.) of changes in tariffs on exports is −0.18 (0.15). Standard errors are clustered at the province level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

At the mean value of migrant intensity (0.34), the overall effect of changes in tariffs on exports is −1.24. This means that prefectures with a larger demand for migrants responded more positively to declines in tariffs on exports. Column 2 shows similar results. Columns 3 and 4 use the private-firm share of sales, and there is a negative interaction effect as well. This indicates that prefectures where a larger share of sales came from private firms responded more strongly to the changes in tariffs on exports. Columns 5–8 present similar results when using the income level of the prefecture to proxy for migrant intensity.

Overall, migrant-intensive prefectures responded more intensively to the changes in tariffs on exports, and this heterogeneity reinforces the conclusion that changes in tariffs on exports caused the changes in migration regulations.

VII. Regulation Changes and Migrant Well-Being

In this section, I provide suggestive evidence on the enforcement of regulations. As explained in section III, migration regulations had specific targets: increasing migrant wages, forcing firms to sign contracts, providing social insurance to migrants, and giving migrant children access to local primary and secondary schools. Thus it would be helpful to see whether the regulations indeed improved these outcomes for migrants and show that these *de jure* regulations were *de facto* effective.

Unfortunately, the only available source that includes these measures is the 2005 1% population survey data. Thus I cannot see how regulation changes affected *changes* in migrant well-being. I can only investigate in the cross-section whether in prefectures with more promigrant regulations, migrants report greater access to local amenities and have higher wages. To alleviate the concern that prefectures with more migrant-friendly regulations could be essentially different from other prefectures, I control for corresponding local worker outcomes, log per capita GDP, and log num-

ber of migrant adults in 2005. Results are shown in table 3 panel A.

Table 3 panel A indicates that the prefectures with higher regulation scores are also the ones with more favorable migrant outcomes, concerning social insurance, income levels, and contract issues. Column 1 shows that a one-standard-deviation increase in the inverse-hyperbolic-sine-transformed regulation score is related to a 0.87-percentage-point increase in the unemployment insurance rate for migrants. Given that the mean insurance rate for migrant is 21% and for locals is 37%, a one-unit increase in the inverse-hyperbolic-sine-transformed regulation score will close 6.25% of the migrant-local gap; columns 2 and 3 show similar patterns for pension and medical insurance. The coefficients are statistically significant at the 5% level. Column 4 shows an insignificant effect of regulation scores on the length of contracts. Column 5 indicates that a one-unit increase in the inverse-hyperbolic-sine-transformed regulation score is related to an income increase of 29.27 yuan per month, which is 50% of the wage gap between locals and migrants. Column 6 shows that the regulation score has no significant impact on school enrollment rates among migrant children at the school age. Column 7 is about whether an increase in regulation scores is correlated with more migrant children brought to prefectures where their parents are working; the result is insignificant.

Panel B regresses local worker outcomes on the regulation score, controlling for the local population size and GDP. Column 4 suggests that a higher regulation score is correlated with a longer terms of contracts for locals. These results might capture the fact that higher-income prefectures usually provide more amenities for both locals and migrants. Column 6 shows a small positive relationship between the regulation score and the local kids' enrollment rate.²⁴

²⁴Given that the baseline enrollment rate is already very high (95%), the increase is very small.

TABLE 3.—THE RELATIONSHIP BETWEEN REGULATION SCORES AND MIGRANT OUTCOMES IN 2005

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A.	Dependent variable: Well-being of migrants, 2005						
	Unemploy. Insurance (rate) (1)	Pension (rate) (2)	Medical Insurance (rate) (3)	Terms of Contract (months) (4)	Monthly Income (yuan) (5)	School Enrollment (rate) (6)	Log # of Children (7)
$\sinh^{-1}$ (regulation score), 2005	0.01* (0.01)	0.02** (0.01)	0.02** (0.01)	0.09 (0.10)	29.27** (13.98)	0.00 (0.00)	−0.02 (0.01)
Well-being of local, 2005	0.57*** (0.09)	0.44*** (0.08)	0.40*** (0.05)	0.54*** (0.04)	0.37*** (0.11)	0.53*** (0.13)	0.10 (0.06)
Observations	246	246	246	246	246	246	246
R-squared	0.43	0.27	0.30	0.33	0.62	0.14	0.96
Panel B.	Dependent variable: Well-being of local population, 2005						
$\sinh^{-1}$ (regulation score), 2005	0.011 (0.011)	0.001 (0.008)	−0.001 (0.010)	0.386** (0.167)	16.553 (17.775)	0.005** (0.002)	−0.047* (0.027)
Observations	246	246	246	246	246	246	246
R-squared	0.43	0.45	0.43	0.22	0.57	0.095	0.90
Mean (s.d.) Y for migrant	.21 (.11)	.34 (.12)	.36 (.12)	4.57 (2.94)	923 (189)	.94 (.04)	4.48 (.99)
Mean (s.d.) Y for local	.37 (.16)	.60 (.14)	.59 (.15)	5.05 (2.88)	981 (336)	.95 (.02)	6.81 (.76)

Panel A regresses migrant well-being measures on the inverse-hyperbolic-sine-transformed regulation score and corresponding well-being measures of the local population, controlling for the log number of adult migrants and the log GDP p.c. in 2005. Panel B regresses local outcomes on the inverse-hyperbolic-sine-transformed regulation score, controlling for the log number of adult locals and the log GDP p.c. in 2005. All regressions are weighted by the 2005 prefecture population. The sample restriction is the same as in table 1 column 5, and two prefectures are not included since their GDP measures are absent in the 2005 *Prefecture Statistics Yearbook*. The mean (s.d.) of $\sinh^{-1}$ (regulation score), in 2005 is 0.41 (0.87). Standard errors are clustered at the province level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Column 7 shows that a larger number of local children is correlated with less generous migration regulations, suggesting potential congestion forces regarding education resources. It is reassuring that columns 1, 2, 3, and 5 do not show significant effects of regulations on local well-being measures. This indicates that regulation effects are not merely reflections of local socioeconomic levels that could affect migrant well-being directly but also the actual improvement through implementation and enforcement of the migrant-related regulations.

Overall, the results in table 3 show that prefectures with higher regulation scores also had better migrant well-being, although the size of the estimates varies by the metric of well-being measure. The significant effects concentrated on unemployment insurance, pension, medical insurance, and wages, and all these aspects were the focus of many migration regulations. These results suggest that more promigrant regulations were associated with improvements in the well-being of migrants in the workplace. The outcomes related to migrant children were not significantly affected by the regulations, and there are several potential explanations. First, school capacities were limited, and it was very costly for prefecture governments to expand the capacity in the short run. Second, prefecture governments may have only wanted the migrant workforce and were reluctant to make substantial changes to incentivize migrant workers to settle down with their family. Third, migrant workers might have viewed their migration as temporary and thus did not want to bring their family, especially considering the fact that migrant children are still not allowed to take the college entrance examination outside their Hukou location.

VIII. The Impact of Trade and Migration Policy Changes on Migrant Flows

The previous sections documented significant changes in migration regulations in response to trade liberalization, and that the regulation changes were associated with improvement of migrant well-being. In this section, I present supplementary analyses asking whether the changes in tariffs and the liberalization-induced relaxation in migration regulations led to large changes in migration flows.

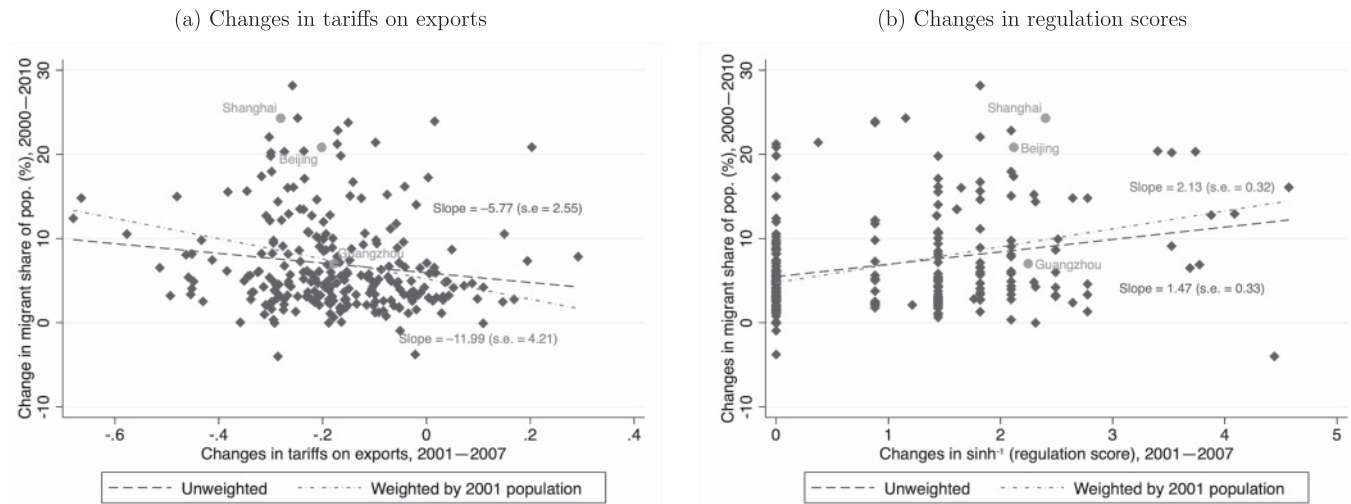
In figure 7(a), the horizontal axis depicts the changes in tariffs on exports from 2001 to 2007, the vertical axis depicts the change in the migrant share of population from 2000 to 2010, and each dot represents a prefecture. The information about the number of migrants is from the 2000 and 2010 censuses. A person is defined as a migrant if he or she has been living in a place other than the Hukou registration place for more than six months or has left the Hukou registration location for more than six months. The graph shows that larger declines in tariffs on exports led to the larger increases in the migrant share. The mega-prefectures Beijing, Shanghai, and Guangzhou are not outliers.²⁵ Panel b depicts the change in the inverse-hyperbolic-sine-transformed regulation score from 2001 to 2007 on the horizontal axis instead, and it shows that the more relaxed the regulation on migrants, the larger the increase in the migrant share.

I estimate reduced-form overall effects of changes in tariffs on exports on economic outcomes by using the following regression equation:

$$\Delta Y_{it} = \gamma_0 + \gamma_1 \Delta \text{tariffs on exports}_{it} + X_{it} \Pi + \xi_{it},$$

²⁵For details of the migrant flow data, refer to appendix B.2.

FIGURE 7.—THE RELATIONSHIP BETWEEN CHANGES IN TARIFFS ON EXPORTS (2001–2007), CHANGES IN THE REGULATION SCORE (2001–2007), AND CHANGES IN THE MIGRANT SHARE OF POPULATION (2000–2010), 248 PREFECTURES



Panel a shows the relationship between the prefecture-level changes in tariffs on exports from 2001 to 2007 and the change in the migrant share of population from 2000 to 2010. Panel b shows the relationship between the change in the inverse-hyperbolic-sine-transformed regulation score from 2001 to 2007 and the change in the migrant share of population from 2000 to 2010. The dashed line is the linear fitted linear, unweighted, and the dotted line is the linear fitted line weighted by the 2001 population. The sample includes 248 prefectures with GDP, wage, and population data from the 2001 *Prefecture Statistics Yearbook*.

where ΔY_{it} can be the 2000–2010 change of the migrant share of population in prefecture i or the change in the log migrant stock. Parameter γ_1 will capture the reduced-form effect of changes in tariffs on exports on outcome variables, including both the direct price channel and the indirect regulation channel. To identify γ_1 , there should be no omitted variable that is correlated with the changes in tariffs on exports and affects the economic outcomes directly. The discussion in section VIA shows that the prefecture-level post-WTO changes in tariffs on exports are not correlated with pre-WTO wage and GDP growth. Thus the identification assumption is likely to be satisfied.

To estimate the effect of regulation changes on migrant flows, I use the following equation,

$$\Delta Y_{it} = \pi_0 + \pi_1 \Delta \sinh^{-1}(\text{regulation score})_{it} + X_{it} \Phi + \zeta_{it},$$

where π_1 represents how regulation changes affect the outcome variables. If π_1 is identified, combining π_1 with α_1 in section VIB, the effect of changes in tariffs on exports on outcome variables through the regulation channel will be $\pi_1 \cdot \alpha_1$. The regulation effect as a share of the total trade effect is $\frac{\pi_1 \cdot \alpha_1}{\gamma_1}$. However, it is challenging to identify the causal impact of the regulation changes on the migrant flows, since the changes in tariffs on exports affect the migrant flows directly, through prices, and indirectly, through the regulation changes, and potentially through other channels. In addition, there could be reverse causality issues since increased migrant flows could motivate regulation changes. Thus the empirically estimated π_1 cannot be interpreted causally.²⁶

Table 4 presents reduced-form effects of changes in tariffs on exports on migrant flows and the association between regulation changes and migrant flows. Panel A column 1 uses changes in the migrant share of the population from 2000 to 2010 as the outcome variable, and the main regressor is the change in tariffs on exports. I also control for the changes in tariffs on imports, the changes in tariffs on intermediate inputs, the migrant share of population in 2000, and the log of population in 2000. Column 1 shows that a one-percentage-point larger decline in tariffs on exports results in a 7.11-percentage-point larger increase in the migrant share of the population. Given the median size of prefecture population in 2001 (3.6 million), the prefectures with a one-standard-deviation larger decline in tariffs on exports had a 38,394 greater increase in the number of migrants. Column 2 shows that a one-unit larger increase in the inverse-hyperbolic-sine-transformed regulation score from 2001 to 2007 was associated with a 1.03-percentage-point larger increase in the migrant share of population. I then add both changes in tariffs on exports and regulation changes together in column 3. Both the coefficient of the changes in tariffs on exports and the coefficient of the regulation change become smaller, suggesting that part of the effect of trade on migrant flows was through the regulation changes.

Now I look at how migrants travel over various distances in response to changes in tariffs on exports and regulation changes in panel A columns 4–6 and panel B. A migrant is defined as a short-distance migrant if he moves within a prefecture; between-prefecture-within-province migrants as medium-distance; and between-province migrants as long-distance. Both changes in tariffs on exports and regulation changes contributed positively to the increase in the number of short-distance migrants. Given the median size of the short-distance migrant population in 2000 (167,000),

²⁶See appendix C.11 for details of the identification challenges and several exercises attempting to address the reverse causality and omitted variable bias issues.

TABLE 4.—THE RELATIONSHIP BETWEEN CHANGES IN TARIFFS ON EXPORTS (2001–2007), REGULATION CHANGES (2001–2007) AND CHANGES IN MIGRANT FLOWS (2000–2010)

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Dependent variable:	Δ migrant share of population			Δ log # of migrants, short-dist.		
Δ tariffs on exports, 2001–2007	–7.11** (3.39)		–6.23* (3.22)	–1.34*** (0.31)		–1.18*** (0.27)
$\Delta \sinh^{-1}$ (regulation score), 2001–2007		1.03*** (0.29)	0.87*** (0.22)		0.18*** (0.04)	0.20*** (0.04)
Observations	248	248	248	238	238	238
R-squared	0.14	0.12	0.16	0.19	0.16	0.25
Mean (s.d.) of depent.		6.94 (5.77)			–0.84 (0.81)	
Panel B. Dependent variable:	Δ log # of migrants, medium-dist.			Δ log # of migrants, long-dist.		
Δ tariffs on exports, 2001–2007	–0.44** (0.20)		–0.39** (0.17)	0.09 (0.39)		0.16 (0.39)
$\Delta \sinh^{-1}$ (regulation score), 2001–2007		0.06* (0.03)	0.05* (0.03)		0.09** (0.04)	0.08* (0.04)
Observations	248	248	248	247	247	247
R-squared	0.59	0.57	0.59	0.12	0.08	0.13
Mean (s.d.) of depent.		1.60 (0.67)			0.85 (0.65)	

Dependent variables are changes from 2000 to 2010. A migrant is defined as a short-distance migrant if he moves within a prefecture; between-prefecture-within-province migrants as medium-distance; and between-province migrants as long-distance. All columns control for changes in tariffs on imports and on intermediate inputs, the log total population and the level of the dependent variable in 2000. The sample restriction is the same as in table 1 column 5. The mean (s.d.) of $\Delta \sinh^{-1}$ (regulation score), 2001–2007 is 1.0 (1.1), and the mean (s.d.) of changes in tariffs on exports is –0.18 (0.15). Standard errors are clustered at the province level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

prefectures with a one-standard-deviation larger declines in tariffs on exports had a 33,567 larger increase in the number of migrants. The overall effect of changes in tariffs on exports is significant for medium-distance migrants but insignificant for long-distance migrants. On the other hand, the correlation between regulation changes and migrant flows is larger among the long-distance migrants than among the medium-distance ones. The results suggest that economic conditions affect short- and medium-distance migration, and when it comes to long-distance migration, regulatory forces on amenities might matter more than economic forces.²⁷

IX. Conclusion

Trade is an important force in shaping economic institutions. This paper uses the changes in tariffs on exports that happened after China entered the WTO to study the effects of trade liberalization on labor institutions that regulate internal migration. I collect a novel data on the prefecture-level government regulations on migrant-related issues to document the change in migrant friendliness in response to changes in tariffs on exports. Empirical estimates show that increased export potentials induced Chinese local governments to provide higher amenities for migrants, and that these indirect trade effects are statistically significant and economically sizable.

²⁷See the appendix on additional results on migrant flows. Appendix C.11.4 uses alternative decomposition of migrant flows and shows that work-related migrants and migrants with more than 12 years of education responded more strongly to regulation changes. Appendix C.11.5 investigates emigration instead of immigration and find no significant effect of changes in tariffs on exports and regulation changes on emigration. Appendix C.11.6 shows that a prefecture that is part of a province with a lot of agricultural population has a bigger inflow of medium-distance migrants once the regulation is relaxed.

This paper focuses on the Hukou system, which regulates internal migration in China. However, trade liberalization can affect other types of economic institutions as well. The external force of WTO rules and the pressure of competing with a bigger international market forced Chinese governments to take measures to improve efficiency and increase transparency. Establishing the rule of law not only affects contemporaneous outcomes but also has long-run impacts on the economy. How to measure the effect of trade liberalization on these broader institutional features is left to future study.

In this paper, individual prefectures choose their own amenity levels to manage the size of their migrant labor force. The increase in overall migrant welfare puts pressure on each individual prefecture to increase their amenity level. This competition between prefectures can decrease the fiscal profit of prefecture governments. If we think about the fiscal profit of local governments as economic rents, then the competition is welfare-improving for the economy since rents become smaller. In addition, the flow of people across geographic areas and sectors can reduce the variance of the national wage distribution and improve total productivity. The current paper does not attempt to answer these productivity-related questions; quantifying this potential productivity effect would also be a fruitful avenue for future work.

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